# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – Analytical Problem Solving Questions

| | | | |
|---|---|---|---|
| **Course Code:** | CSA0303 | **Course Title:** | Data Structures |
| **CO Assessed:** | **CO2** | **Assessment:** | Assessment Tool 1 – Analytical Problem Solving |
| **Weightage:** | 10% of CIA | **Total Marks:** | 100 |
| **CO2 Statement:** | ADT array, Operations on arrays, Searching Algorithms, Linear Search, Binary Search, Suffix Arrays & LCP Arrays ADT, Linked List, Stack, Queue, Applications. | | |
| **Student Name:** | M.NANDHINI | **Reg. No.:** | 192472318 |
| **Section / Batch:** | SLOT-C | **Date:** | 30-07-26 |

**Code of Conduct**

*I ____M.NANDHINI(192472318)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____M.NANDHINI_____

**Instructions**

- This test contains 3 Analytical Problem Solving questions. Total: 100 Marks.
- When a student answer using a analytical problem, in evaluation the answer should contain following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- Electronic Gadgets are not permitted in the class. Take it as test and submit in the class.

| Section / Topic | Focus | Q Nos | Marks |
|---|---|---|---|
| Stack Data Structures | Stack – top movements | 1 | 30 |
| Queue Data Structures | Circular Queue | 2 | 40 |
| Application of Stack | Infix to post-fix | 3 | 40 |
| | **GRAND TOTAL:** | **100 Marks** | |

## Annexure A – Analytical Problem Solving

1) Perform the following operations in LIFO data structure having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(),Pop(). Print the peak element of the data structure and its index position. With a neat sketch, explain in detail.

2)Perform the following operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the front and rear elements with its positions. Discuss in detail.

3)Convert the following infix expression into a post fix expression. B - ( C/D + (E % F* G ) / H) * J. Also, evaluate the given postfix expressions. a) 9 3 8 4 /*+ b) 6 2 + 6 3 /*

### Rubrics for Calculation

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Problem Context | 20 | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts | 30 | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy | Fails to apply appropriate concepts |
| Problem-Solving Approach | 20 | Logical, well-structured, and efficient approach | Mostly logical approach with minor gaps | Basic approach with some inconsistencies | Illogical or unclear approach |
| Accuracy of Solution | 20 | Completely correct solution with proper justification | Mostly correct with minor errors | Partially correct solution | Incorrect or incomplete solution |
| Clarity | 10 | Clear, well-organized, and properly explained solution | Understandable with minor clarity issues | Limited clarity and explanation | Poorly presented and difficult to understand |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| | | |
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |

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CSA0303

# CO2-AT1- Test

1 Perform the following operations on LIFO data structure having an array size of 5. Push (45), Push(22), Push(11) Push (77), POP(), Push(44), Push(33), POP(), POP(), POP(), Print the Peak element of the dt and its index position with a neat Sketch explained in detail

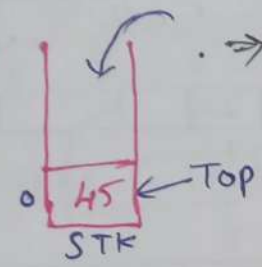
array size = 5

Stack => STK[5]

Operations => Push(45), Push(22), Push(11), Push (77), POP(), Push(44), Push(33), POP(), POP(), POP()

Sketch

Push(45) =>

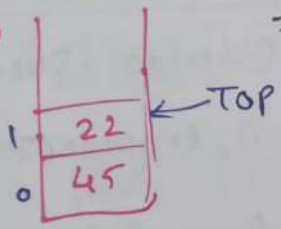



=> Top = -1, Size = 5

Push (45) => STK[TOP] => STK[0] = 45

Top = Top + 1 => -1 + 1 = 0.

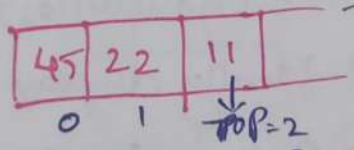
Push (22) =>




=> push (22) => STK[1] => 22

Top = Top + 1 = 0 + 1

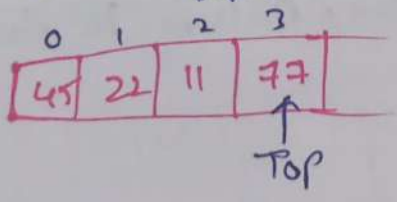
Push (11) =>




=> push (11) => STK[2] => 11

Top = Top + 1 => 1 + 1 = 2

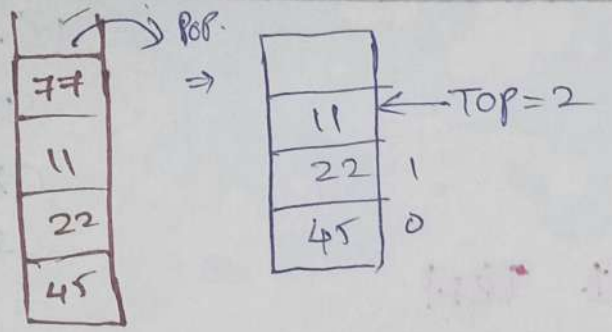
Push (77) =>




=> Push (77) => STK[3] = 77

Top = Top + 1 => 2 + 1 => 3

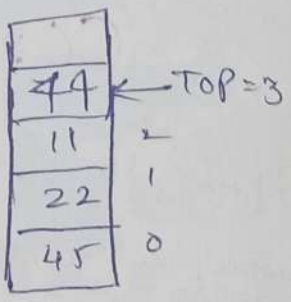
* Pop() ⇒




* Push(44) ⇒ STK[TOP] ⇒ STK[3] ⇒ 44

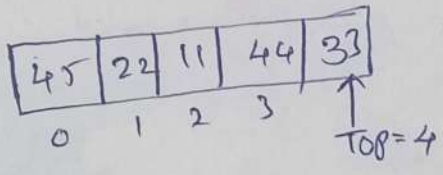
Top = Top+1 ⇒ 2+1 = 3

⇒

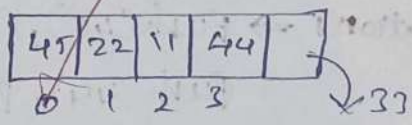



* Push(33) ⇒ STK[TOP] ⇒ STK[4] ⇒ 33

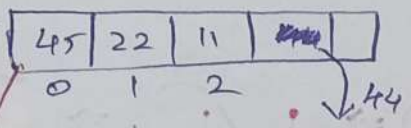
Top = Top+1 ⇒ 3+1 = 4



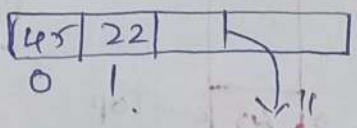

* POP() ⇒ Top = Top - 1 ⇒ 4-1 = 3 ⇒



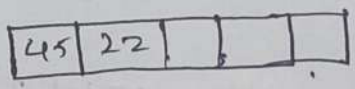

* POP() ⇒ Top = Top - 1 ⇒ 3-1 ⇒ 2 ⇒




* POP() ⇒ Top ⇒ Top - 1 ⇒ 2-1 ⇒ 1 ⇒



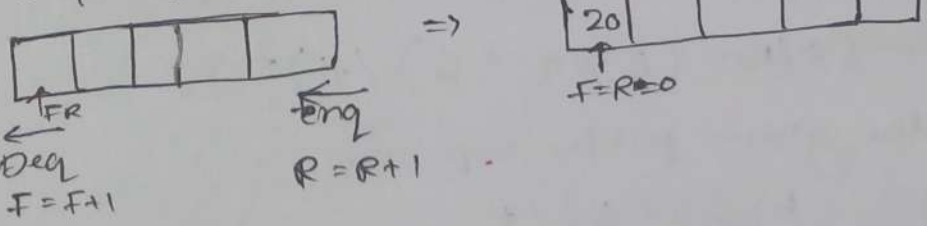

Final Stack ⇒




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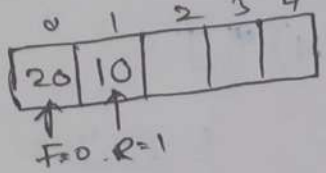
2 Perform the following Operations in Circular Queue having a size of 5. Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90). Finally display the Front most elements with its positions, Discuss in detail.

* Enqueue (20) ⇒



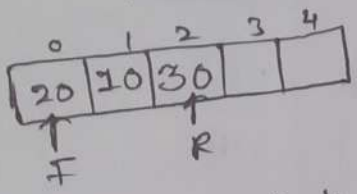

Front = Rear = -1

* Enque (10) ⇒



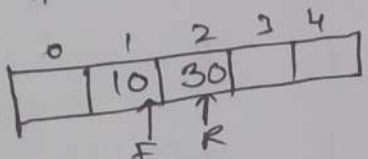

R → R+1

* Enq (30) ⇒



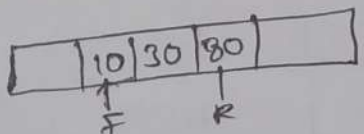

R → R + 1

* Deq (·) ⇒



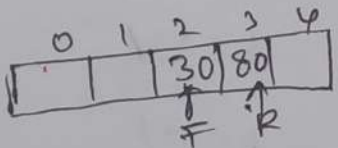

F → F+1

* Enque (80) ⇒



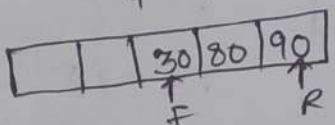

R → R+1

* Deq ( ) ⇒



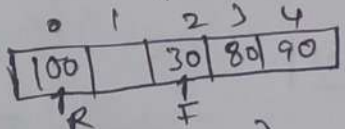

R → R+1

* Enq (90) ⇒

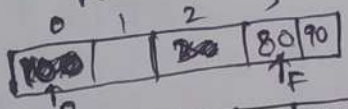



⇒ R → R+1

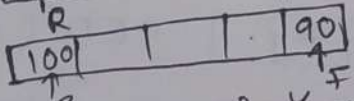
* Enq (100) ⇒



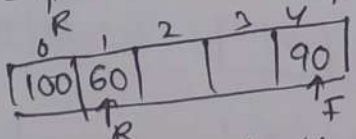

* Deq ( ) ⇒



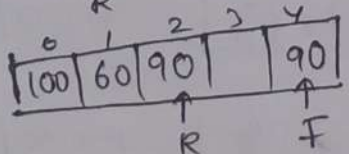

* Deq ( ) ⇒




* Enq (60) ⇒




* Enq (90) ⇒




⇒ Final queue

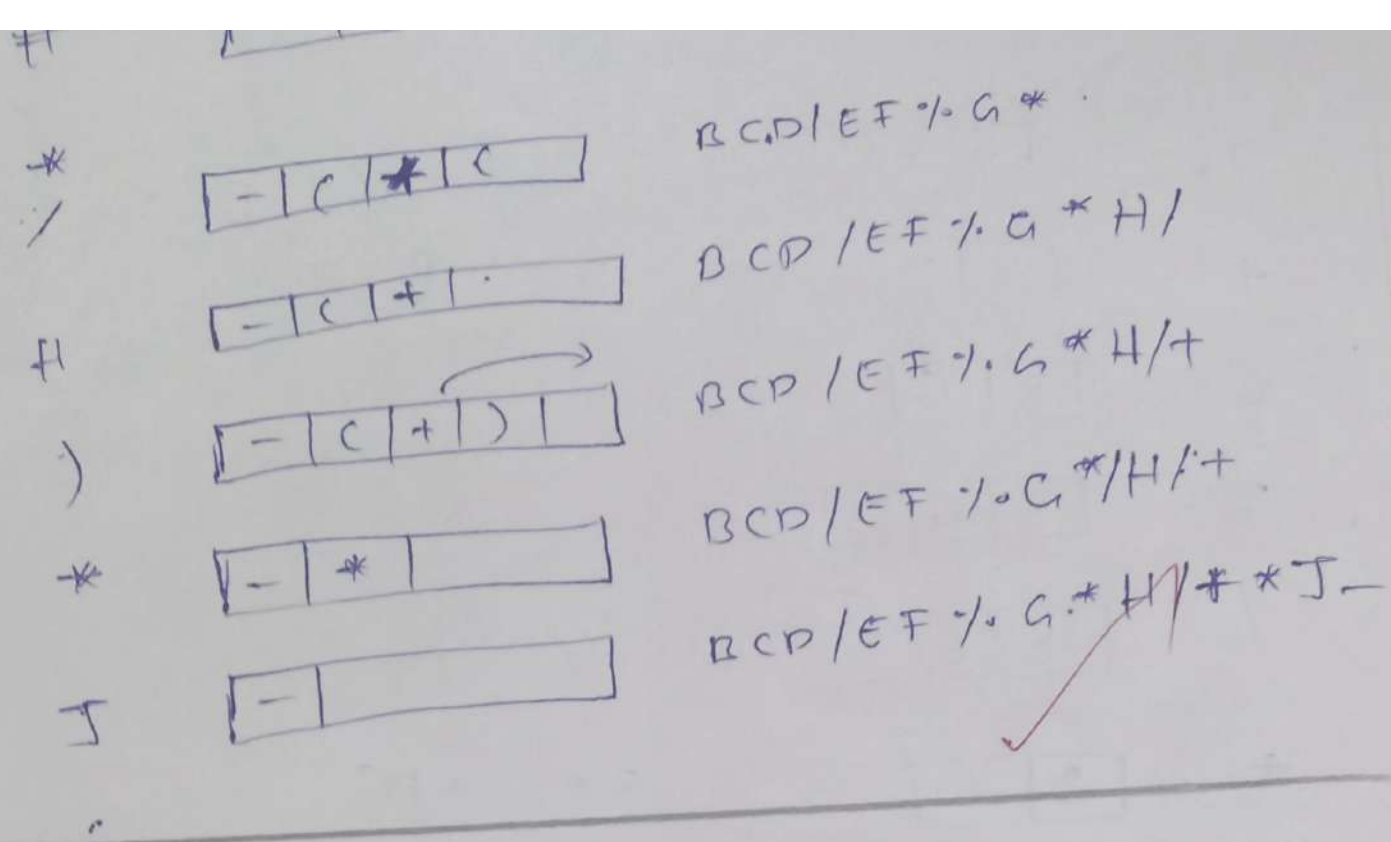
3 Convert the following infix expression into a post fix expression. ⇒ B − (C/D + (E % F * G) / H) * J.

Also evaluate the given postfix expression

a) 9 3 8 4 / * +  b) 6 2 + 6 3 / *.

Sol B − (C/D + (E % F * G) / H) * J

| Scanning char | Stack | Expression |
|---|---|---|
| B | | B |
| − | − | B |
| ( | − ( | B |
| C | − ( | BC |
| / | − ( / | BC |
| D | − ( / | BCD |
| + | − ( + | BCD/ |
| ( | − ( + ( | BCD/ |
| E | − ( + ( | BCD/E |
| % | − ( + ( % | BCD/E |
| F | − ( + ( % | BCD/EF |
| * | − ( + ( * | BCD/EF% |
| G | − ( + ( * | BCD/EF%G |
| ) | − ( + ( * ) | BCD/EF%G* |
| / | − ( + / | BCD/EF%G*H |

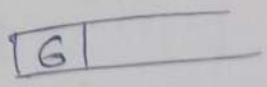
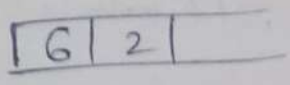
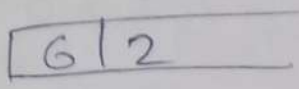
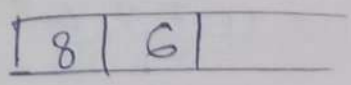
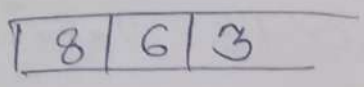
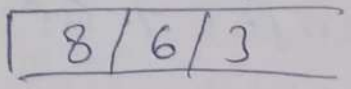
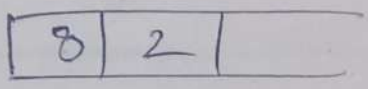



a) 9 3 8 4 / * +

| Scanning | Stack | Expn |
|---|---|---|
| 9 | 9 | |
| 3 | 9 3 | |
| 8 | 9 3 8 | |
| 4 | 9 3 8 4 | |
| / | 9 3 8 4 | 8/4 = 2 |
| * | 9 3 2 | 3 * 2 = 6 |
| + | 9 6 | 9 + 6 ⇒ 15 |

b) 6 2 + 6 3 / *

Expression

| Scanning | Stack | |
|---|---|---|
| 6 |  | |
| 2 |  | |
| + |  | 6+2=8 |
| 6 |  | |
| 3 |  | |
| / |  | 6/3=2 |
| * |  | 8*2 ⇒ 16. |